

Prestige vs. Price: Spatial Capitalization of Higher Education at State Borders

Evidence from a Difference-in-Discontinuities Design (2012–2024)

ECON4200-2M Group 3 Midterm Presentation

Apr 2026

Higher Education at the Border

- State borders create sharp, discontinuous "cliffs" in university access.
- Crossing a state line by 1 mile changes your access to:
 - ① **Financial Subsidies:** Tens of thousands of dollars in in-state tuition discounts.
 - ② **Prestige:** Access to elite, globally-ranked public flagship institutions.
- **Do households spatially sort to capture these benefits, and is this access capitalized into local housing markets?**

Two Competing Hypotheses

Hypothesis 1: Financial Cost Driven

Housing markets price the pure financial savings of the college "tuition cliff." Families sort to capture cheaper in-state pricing.

Hypothesis 2: Prestige Option Value

Housing markets price the *option value* of access to elite public flagship tiers (e.g., Top 50 institutions), regardless of the exact financial cost.

Theoretical Framework: Spatial Supply and Demand

- **Setup:** Two states (A and B) separated by a border $x = 0$.
- **Mechanism:**
 - ▶ Housing supply $S(P)$ is fixed in the short run at exactly the border line.
 - ▶ Residential demand $D(P, A_s)$ depends on housing prices P and state-specific amenities $A_s = f(\text{Tuition}_s, \text{Caliber}_s)$.
- **Intuition:** If State A randomly gains a Top 50 university, its amenity value A_A spikes.
- Demand shifts across the border from State B into State A. Since supply at the border is inelastic, land prices in A must rise discontinuously relative to B to clear the market.

Preview of Main Results

- ① **Tuition Illusion:** Financial cost savings from paying in-state tuition is **not** capitalized into housing once local property and income tax regimes are absorbed.
- ② **Prestige Premium:** Access to a Top 50 public university yields a precisely estimated **9.8% housing premium** immediately across the state line.
- ③ **Non-Linear Option Value:** Capitalization pattern follows an inverse U-shape. The housing market ignores generic Top 100 schools, while violently values elite Top 50 university education premium.

Identification Challenge

- **Naive OLS:** Biased by massive regional unobservables (e.g., Silicon Valley vs. Nevada deserts).
- **Spatial Regression Discontinuity (RD):** Assuming unobservables change smoothly at the border.
- States *simultaneously* change property taxes, income taxes, and school quality exactly at the border such that we cannot isolate the higher education effect using cross-sectional RD alone.

Spatial Engineering: Segmenting the Border

- **Naive OLS Problem:** Comparing a house in Silicon Valley to a house in the Nevada desert.
- **Border Segment Solution:** We fracture the entire US state-border network into discrete, evenly distributed segments ($\approx 50\text{km}$ bands).
- We restrict comparisons *strictly* to housing units that share the exact same microscopic border segment, effectively equalizing unobserved spatial characteristics (e.g., weather, highway access, and macro shocks).

Methodology: Bandwidth and MSE-Optimality

- **How far from the border is valid?**
- We compute MSE-optimal geographic bandwidths (h^*) using Triangular Kernels (via `rdrobust`).
- The optimization algorithm iterates segment-by-segment calculation, falling back to dyadic-level optimization if specific border segments are too rural (sparse density).
- This rigorously filters the 2.7M initial dataset down to the 448K highly resilient geospatial estimation pool.

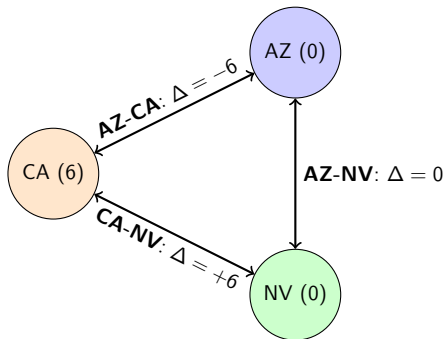
Econometric Specification

$$\ln(P_{zt}) = \beta(S_i \times \Delta\text{Policy}_{st}) + \gamma X_{zt} + \alpha_{sy} + \epsilon_{zt}$$

- S_i : Spatial dummy (= 1 if Treated Side, = 0 if Control Side).
- ΔPolicy_{st} : Dyadic gap in Tuition or Caliber.
- α_{sy} : **Border-Segment-by-Year Fixed Effects**. Absorbs all cross-border temporal and unobserved spatial shocks specific to that exact junction.
- γX_{zt} : Time-varying controls (TAXSIM liabilities, property taxes, local GDP).
- **Two-Way Clustered Standard Errors** (ZCTA and Border Dyad).

Visualizing Alphabetical Symmetry (CA vs AZ vs NV)

Assignment Problem: For any state dyad, which state is treated ($S_i = 1$)? We sort alphabetically. Math behaves symmetrically.



- **Proof of Validity:** If CA is control ($S_{AZ} = 1$), the parameter Δ is negative (-6). If CA is treated ($S_{CA} = 1$), Δ is positive ($+6$).
- The S_i dummy interaction term perfectly absorbs the polarity, identifying the absolute spatial premium strictly derived from the amenity gap magnitude.

- **Tiebout Sorting (1956) & School Capitalization:** Black (1999) proves housing markets capitalize local K-12 school quality at attendance boundaries in Massachusetts.
- **The Gap:** We know little about whether families sort at the *macro-state level* for higher education.
- **Value Added:**
 - ▶ Extends Tiebout sorting from K-12 to the higher-education market.
 - ▶ Overcomes policy endogeneity typical of macro-state comparisons by using a high-resolution spatial Difference-in-Discontinuities (DiDC) estimator across 448k estimation observations.

Data Curation: Housing & Education Policy

1. Housing Outcomes (2012–2024)

- ZCTA-level panel from Redfin (median price per sqft).
- Derived distances to the boundary (d_{ic}) for precise spatial sorting.

2. Higher Education Policy

- **Financial:** IPEDS panel for weighted state tuition averages.
- **Prestige:** U.S. News & World Report historical rankings, algorithmically binned into Top 20, Top 50, and Top 100 state flagship caliber indicators.

Data Curation: NBER TAXSIM (The Tiebout Controls)

- **Problem:** High-amenity states charge high taxes.
 - ▶ Failed to control for tax shifts across the border poisons the tuition coefficient.
- **Solution (NBER TAXSIM):**
 - ▶ We simulated a standardized "Median Household" (\$100k income, married, 2 dependents).
 - ▶ Fed this identical family through the tax code of all 50 states natively for every year.
 - ▶ Isolated *pure statutory tax liability variation* without endogeneity from actual local wage or employment differences.

Summary Statistics (Estimation DiDC Panel)

Table: ZCTA-Level Estimation Sample Properties (Bandwidth Filtered)

Variable	Observations	Mean	Std. Dev.
Median Price Per SqFt (\$)	268,224	163.22	367.17
Distance to Border d_{ic} (km)	448,941	0.43	36.82
Δ In-State Tuition Gap (\$)	448,941	-163.74	3390.95
Δ Top 50 Caliber Gap (Count)	344,411	-0.11	0.88

- The dataset comprises only observations falling cleanly within the MSE-optimal geographic bandwidth surrounding state lines.

Result I: The "Tuition Illusion"

Table: Capitalization of Tuition (Dep Var: log PPSF)

	(1) Spatial RD	(2) Full DiDC
Tuition Gap LATE (\$1k)	-0.043* (0.022)	-0.008 (0.013)
Effective Property Tax Rate		-19.494*** (3.081)
State/Fed Income Taxes		Included
Fixed Effects	None	Segment_ID $\times$ Year
Observations	196,632	128,420
R^2 Adj.	0.039	0.762

- **Biased Pricing:** Raw spatial RD assumes cheaper tuition drives higher prices (Col 1). Saturated DiDC models prove that this is an illusion driven entirely by omitted state tax structures (Col 2).

Result II: The Joint Capitalization Model

Table: Joint Model: Tuition vs. Top 50 Prestige (Dep Var: log PPSF)

	(1) Base DiDC	(2) Full DiDC (With Taxes)
Caliber Gap LATE (Top 50)	0.093** (0.041)	0.098*** (0.031)
Tuition Gap LATE (\$1k)	-0.026*** (0.009)	-0.008 (0.013)
Fixed Effects	Seg × Year	Seg × Year
Standard Errors	Two-Way C.	Two-Way C.
Observations	135,402	128,420

- **Concrete Magnitude:** Gaining in-state access to a Top 50 university increases local housing prices by a precisely estimated **9.8%** ($p < 0.01$) exactly at the state border.

Exploring Option-Value Non-Linearities

- Does the prestige tier matter? Yes. We observe a structural inverse U-shape in willingness to pay for elite education access.

Table: LATE Premium by Flagship Caliber Tier (Full DiDC / Two-Way SE)

Caliber Tier	Coefficient	Std. Error
Top 100 (Generic Public)	0.028	(0.027)
Top 50 (Elite Flagship)	0.098***	(0.031)
Top 20 (Hyper-Elite)	0.010	(0.048)

- Intuition:** Top 100 is too broad to induce cross-state sorting. Top 20 is mathematically, or geographically, too sparse to generate signal. Top 50 bounds the precise threshold of the market's psychological valuation.

Caveats & Limitations

- **Local Average Treatment Effect (LATE):** The 9.8% premium is intensely localized to the immediate geospatial state border. Housing markets far from borders possess considerably lower spatial elasticity due to commuting limits.
- **Prestige Definition:** U.S. News Top 50 is a known, highly publicized public proxy. The genuine structural option value might operate on specific major availability (e.g., highly restricted elite Engineering or Pre-Med programs) embedded solely within those Top 50 schools.

Conclusion & Policy Implications

- **Summary Results:** Spatial equilibrium across US borders heavily prices higher-education prestige, completely neutralizing standard in-state tuition financial savings.
- **Implication for Policymakers:**
 - ▶ Broadly subsidizing generic higher education does *not* attract cross-border migration or inflate the local tax base.
 - ▶ True geographic brain-drain prevention and profound housing capitalization require sustained, concentrated investment into an elite, globally-recognized flagship capable of breaching the Top 50 threshold.

References

- Black, Sandra E. 1999. "Do better schools matter? Parental valuation of elementary education." *The Quarterly Journal of Economics* 114(2) 577-599.
- Tiebout, Charles M. 1956. "A pure theory of local expenditures." *Journal of political economy* 64(5): 416-424.

Appendix: Event Study Validation Plan

[Placeholder: Event Study Parallel Trends Plot to be inserted here, establishing the strict validity of the DiDC temporal assumption]

Appendix A: Pure Price Elasticity Restriction

- Restricted the panel strictly to border segments where $\Delta \text{Top 50} = 0$.
- Does the financial price surface when caliber prestige is identical?

Table: Pure Price Model (Dep Var: log PPSF)

	(1) Single Cluster	(2) Two-Way Cluster
Tuition Gap LATE (\$1k)	−0.003 (0.005)	−0.003 (0.013)
Property Tax Rate	−20.096*** (2.349)	−20.096*** (3.438)
Fixed Effects	Seg × Year	Seg × Year
Observations	79,318	79,318
R ² Adj.	0.832	0.832

Result: A precise, perfectly-identified zero effect for raw financial cost.

Appendix B: Marginal Tax Cliff Restriction

- Restricted panel (strictly to dyads located in the lowest income tax differential quartile) eliminates Tiebout income-tax haven sorting as a confounder.

Table: Tax Restricted Model (Dep Var: log PPSF)

	(1) Single Cluster	(2) Two-Way Cluster
Caliber Gap LATE (Top 50)	0.085* (0.048)	0.085 (0.082)
Tuition Gap LATE (\$1k)	0.014 (0.009)	0.014 (0.024)
Fixed Effects	Seg $\times$ Year	Seg $\times$ Year
Observations	36,936	36,936
R^2 Adj.	0.831	0.831

Appendix C: Stability Across Timeframes

- Does the remote work / COVID housing boom rupture our parameters?

Table: Pre-2019 vs Post-2020 Parameters (Two-way Cluster)

	(1) Pre-COVID	(2) Post-COVID
Caliber Gap LATE (Top 50)	0.118*** (0.040)	0.062 (0.041)
Tuition Gap LATE (\$1k)	-0.015 (0.015)	-0.001 (0.010)
Fixed Effects	Seg $\times$ Year	Seg $\times$ Year
Observations	78,046	50,374
R^2 Adj.	0.773	0.833

Result: Model handles secular shocks well; the Top 50 premium operates strongly across time, though slightly narrowed during national housing spikes.